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Method and system for monitoring wireless link quality of handheld ultrasound devices

Abstract

A system and method for providing feedback to a user of a wireless ultrasound probe regarding a wireless link quality between the ultrasound probe and a display device is provided. The method includes receiving, by a display processor of a display device, an acquisition frame rate from a wireless ultrasound probe. The method includes determining, by the display processor, an analysis interval length based on the acquisition frame rate and a wireless connection state loss threshold. The method includes receiving and counting ultrasound image frames wirelessly transmitted to the display device from the wireless ultrasound probe during the analysis interval length. The method includes determining that a number of lost ultrasound frames during the analysis interval length exceeds the wireless connection state loss threshold. The method includes causing a display system of the display device to present a bad connection state identifier.

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Background/Summary

FIELD

(1) Certain embodiments relate to ultrasound imaging. More specifically, certain embodiments relate to a method and system for providing feedback to a user of a wireless ultrasound probe regarding a wireless link quality between the ultrasound probe and a display device.

BACKGROUND

(2) Ultrasound imaging is a medical imaging technique for imaging organs and soft tissues in a human body. Ultrasound imaging uses real time, non-invasive high frequency sound waves to produce a series of two-dimensional (2D) and/or three-dimensional (3D) images.

(3) Wireless ultrasound probes send ultrasound data wirelessly to a display device. The ultrasound data may be transmitted using an unreliable UDP-based protocol. Accordingly, image frames may be lost during transmission or reception of the image frames from the wireless ultrasound probe to the display device, such as if a network socket is not ready to accept more data at a time an image frame is ready to be sent or if one of the UDP packets are dropped in transit. Additionally, image frames may be lost within the wireless ultrasound probe, such as when frame processing of a wireless ultrasound probe is temporarily limited due to high processor load.

(4) Further limitations and disadvantages of conventional and traditional approaches will become apparent to one of skill in the art, through comparison of such systems with some aspects of the present disclosure as set forth in the remainder of the present application with reference to the drawings.

BRIEF SUMMARY

(5) A system and/or method is disclosed for providing feedback to a user of a wireless ultrasound probe regarding a wireless link quality between the ultrasound probe and a display device, substantially as shown in and/or described in connection with at least one of the figures, as set forth more completely in the claims.

(6) These and other advantages, aspects and novel features of the present disclosure, as well as details of an illustrated embodiment thereof, will be more fully understood from the following description and drawings.

Description

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWINGS

(1) FIG. 1 is a block diagram of an exemplary ultrasound system that is operable to provide feedback to a user of a wireless ultrasound probe regarding a wireless link quality between the ultrasound probe and a display device, in accordance with various embodiments.

(2) FIG. 2 illustrates a screenshot of an exemplary ultrasound image display having an indication of

a good connection status, in accordance with various embodiments.

(3) FIG. 3 illustrates a screenshot of an exemplary ultrasound image display having an indication of a bad connection status, in accordance with various embodiments.

(4) FIG. 4 is a flow chart illustrating exemplary steps that may be utilized for providing feedback to a user of a wireless ultrasound probe regarding a wireless link quality between the ultrasound probe and a display device, in accordance with various embodiments.

DETAILED DESCRIPTION

(5) Certain embodiments may be found in a method and system for providing feedback to a user of a wireless ultrasound probe regarding a wireless link quality between the ultrasound probe and a display device. For example, aspects of the present disclosure have the technical effect of providing a display device with an acquisition frame rate of a wireless ultrasound probe. Moreover, aspects of the present disclosure have the technical effect of determining an analysis interval length of received ultrasound image frames based on an acquisition frame rate and a wireless connection state threshold. Furthermore, aspects of the present disclosure have the technical effect of dynamically updating an analysis interval length based on changes to an acquisition frame rate at a wireless ultrasound probe. Additionally, aspects of the present disclosure have the technical effect of presenting a connection state at a display system of display device.

(6) The foregoing summary, as well as the following detailed description of certain embodiments will be better understood when read in conjunction with the appended drawings. To the extent that the figures illustrate diagrams of the functional blocks of various embodiments, the functional blocks are not necessarily indicative of the division between hardware circuitry. Thus, for example, one or more of the functional blocks (e.g., processors or memories) may be implemented in a single piece of hardware (e.g., a general-purpose signal processor or a block of random access memory, hard disk, or the like) or multiple pieces of hardware. Similarly, the programs may be stand alone programs, may be incorporated as subroutines in an operating system, may be functions in an installed software package, and the like. It should be understood that the various embodiments are not limited to the arrangements and instrumentality shown in the drawings. It should also be understood that the embodiments may be combined, or that other embodiments may be utilized, and that structural, logical and electrical changes may be made without departing from the scope of the various embodiments. The following detailed description is, therefore, not to be taken in a limiting sense, and the scope of the present disclosure is defined by the appended claims and their equivalents.

(7) As used herein, an element or step recited in the singular and preceded with the word “a” or “an” should be understood as not excluding plural of said elements or steps, unless such exclusion is explicitly stated. Furthermore, references to “an exemplary embodiment,” “various embodiments,” “certain embodiments,” “a representative embodiment,” and the like are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. Moreover, unless explicitly stated to the contrary, embodiments “comprising”, “including”, or “having” an element or a plurality of elements having a particular property may include additional elements not having that property.

(8) Also as used herein, the term “image” broadly refers to both viewable images and data representing a viewable image. However, many embodiments generate (or are configured to generate) at least one viewable image. In addition, as used herein, the phrase “image” is used to refer to an ultrasound mode such as B-mode (2D mode), three-dimensional (3D) mode, 3D Zoom mode (e.g., thin slab), M-mode, CF-mode, PW Doppler, CW Doppler, Contrast Enhanced Ultrasound (CEUS), and/or sub-modes of B-mode and/or CF such as Harmonic Imaging, Shear Wave Elasticity Imaging (SWEI), Strain Elastography, TVI, PDI, B-flow, MVI, UGAP, and in some cases also MM, CM, TVD where the “image” and/or “plane” includes a single beam or multiple beams.

(9) Furthermore, the term processor or processing unit, as used herein, refers to any type of

processing unit that can carry out the required calculations needed for the various embodiments, such as single or multi-core: CPU, Accelerated Processing Unit (APU), Graphic Processing Unit (GPU), DSP, FPGA, ASIC or a combination thereof.

(10) It should be noted that various embodiments described herein that generate or form images may include processing for forming images that in some embodiments includes beamforming and in other embodiments does not include beamforming. For example, an image can be formed without beamforming, such as by multiplying the matrix of demodulated data by a matrix of coefficients so that the product is the image, and wherein the process does not form any “beams”. Also, forming of images may be performed using channel combinations that may originate from more than one transmit event (e.g., synthetic aperture techniques).

(11) In various embodiments, ultrasound processing to form images is performed, for example, including ultrasound beamforming, such as receive beamforming, in software, firmware, hardware, or a combination thereof. One implementation of an ultrasound system having a software beamformer architecture formed in accordance with various embodiments is illustrated in FIG. 1.

(12) FIG. 1 is a block diagram of an exemplary ultrasound system **100** that is operable to provide feedback to a user of a wireless ultrasound probe **104** regarding a wireless link quality between the ultrasound probe **104** and a display device **150**, in accordance with various embodiments. Referring to FIG. 1, there is shown an ultrasound system **100** comprising a wireless ultrasound probe **104** and a display device **150**. The wireless ultrasound probe **104** comprises a transmitter **102**, a transmit beamformer **110**, a receiver **118**, a receive beamformer **120**, transducer elements **106**, transmit **114** and receive **116** sub-aperture beamformers, A/D converters **122**, a RF processor **124**, a RF/IQ buffer **126**, a probe signal processor **132**, and a probe transceiver.

(13) The transmitter **102** of the wireless ultrasound probe **104** may comprise suitable logic, circuitry, interfaces and/or code that may be operable to drive the wireless ultrasound probe **104**. The wireless ultrasound probe **104** may comprise a two-dimensional (2D) array of piezoelectric elements. In various embodiments, the wireless ultrasound probe **104** may comprise a matrix array transducer or any suitable transducer operable to acquire 2D and/or 3D (including 4D) ultrasound image datasets. The wireless ultrasound probe **104** may comprise a group of transmit transducer elements **106** and a group of receive transducer elements **108**, that normally constitute the same elements. In certain embodiment, the wireless ultrasound probe **104** may be operable to acquire ultrasound image data covering at least a substantial portion of an anatomy, such as a heart, a fetus, a lung, a blood vessel, or any suitable anatomical structure.

(14) The transmit beamformer **110** may comprise suitable logic, circuitry, interfaces and/or code that may be operable to control the transmitter **102** which, through a transmit sub-aperture beamformer **114**, drives the group of transmit transducer elements **106** to emit ultrasonic transmit signals into a region of interest (e.g., human, animal, underground cavity, physical structure and the like). The transmitted ultrasonic signals may be back-scattered from structures in the object of interest, like blood cells or tissue, to produce echoes. The echoes are received by the receive transducer elements **108**.

(15) The group of receive transducer elements **108** in the wireless ultrasound probe **104** may be operable to convert the received echoes into analog signals, undergo sub-aperture beamforming by a receive sub-aperture beamformer **116** and are then communicated to a receiver **118**. The receiver **118** may comprise suitable logic, circuitry, interfaces and/or code that may be operable to receive the signals from the receive sub-aperture beamformer **116**. The analog signals may be communicated to one or a plurality of A/D converters **122**.

(16) The plurality of A/D converters **122** may comprise suitable logic, circuitry, interfaces and/or code that may be operable to convert the analog signals from the receiver **118** to corresponding digital signals. The plurality of A/D converters **122** are disposed between the receiver **118** and the RF processor **124**. Notwithstanding, the disclosure is not limited in this regard. Accordingly, in some embodiments, the plurality of A/D converters **122** may be integrated within the receiver **118**.

(17) The RF processor **124** may comprise suitable logic, circuitry, interfaces and/or code that may be operable to demodulate the digital signals output by the plurality of A/D converters **122**. In accordance with an embodiment, the RF processor **124** may comprise a complex demodulator (not shown) that is operable to demodulate the digital signals to form I/Q data pairs that are representative of the corresponding echo signals. The RF or I/Q signal data may then be communicated to an RF/IQ buffer **126**. The RF/IQ buffer **126** may comprise suitable logic, circuitry, interfaces and/or code that may be operable to provide temporary storage of the RF or I/Q signal data, which is generated by the RF processor **124**.

(18) The receive beamformer **120** may comprise suitable logic, circuitry, interfaces and/or code that may be operable to perform digital beamforming processing to, for example, sum the delayed channel signals received from RF processor **124** via the RF/IQ buffer **126** and output a beam summed signal. The resulting processed information may be the beam summed signal that is output from the receive beamformer **120** and communicated to the probe signal processor **132**. In accordance with some embodiments, the receiver **118**, the plurality of A/D converters **122**, the RF processor **124**, and the beamformer **120** may be integrated into a single beamformer, which may be digital. In various embodiments, the ultrasound system **100** comprises a plurality of receive beamformers **120**.

(19) The probe signal processor **132** may comprise suitable logic, circuitry, interfaces and/or code that may be operable to process ultrasound scan data (i.e., summed IQ signal) for generating ultrasound image frames for transmission via a transceiver (not shown) to the display device **150**. The probe signal processor **132** is operable to perform one or more processing operations according to a plurality of selectable ultrasound modalities on the acquired ultrasound scan data. Acquired ultrasound scan data may be processed in real-time during a scanning session as the echo signals are received. Additionally or alternatively, the ultrasound scan data may be stored temporarily in the RF/IQ buffer **126** during a scanning session and processed in less than real-time in a live or off-line operation.

(20) The probe signal processor **132** may be one or more central processing units, graphic processing units, microprocessors, microcontrollers, and/or the like. The probe signal processor **132** may comprise an acquisition frame rate processor **140** and may be capable of receiving input information from the display device **150**, communicating an acquisition frame rate to the display device **150**, and transmitting ultrasound image frames to the display device **150**, among other things. The probe signal processor **132** and acquisition frame rate processor **140** may be capable of executing any of the method(s) and/or set(s) of instructions discussed herein in accordance with the various embodiments, for example.

(21) The ultrasound system **100** may be operable to continuously acquire ultrasound scan data at an acquisition frame rate that is suitable for the imaging situation in question. Typical acquisition frame rates range from 5-30 frames per second but may be lower or higher. The acquired ultrasound scan data processed as ultrasound image frames by the probe signal processor **132** may be wirelessly transmitted to the display device **150** for further display processing and display on the display system **134** at a display-rate that can be the same as the frame rate, or slower or faster. The wireless connection between the wireless probe **104** and the display device **150** may be Wi-Fi or any suitable wireless connection. In various embodiments, a probe transceiver may be configured to transmit ultrasound image frames from the probe signal processor **132** to a display device transceiver of the display device **150** via the wireless connection according to the user datagram protocol (UDP) communications protocol. In certain embodiments, the probe transceiver may be configured to transmit the acquisition frame rate and other suitable messages from the acquisition frame rate processor **140** and/or the probe signal processor **132** via the wireless connection to the display device transceiver of the display device **150** according to the transmission control protocol (TCP) communications protocol. In a representative embodiment, the probe transceiver may be configured to receive messages from the display device transceiver of the display device **150** via

the wireless connection according to the transmission control protocol (TCP) communications protocol. For example, the probe transceiver may receive messages related to an imaging mode, imaging parameters, imaging presets, settings, and/or the like from a display device signal processor **152** of the display device **150** via the display device transceiver. The messages may be provided by the probe transceiver to the probe signal processor **132** for controlling configuration and/or operation of the wireless ultrasound probe **104**.

(22) The probe signal processor **132** may include an acquisition frame rate processor **140** that comprises suitable logic, circuitry, interfaces and/or code that may be operable to determine an acquisition frame rate of the wireless ultrasound probe **104**. For example, the probe signal processor **132** may receive an imaging mode, imaging parameters, imaging presets, settings, and/or the like from the display device **150** via the wireless connection (e.g., using TCP) to configure the wireless ultrasound probe **104** for ultrasound image acquisition. The acquisition frame rate processor **140** may be configured to determine the acquisition frame rate based on the configuration and operating condition of the wireless ultrasound probe **104**. The acquisition frame rate processor **140** may be configured to transmit the determined acquisition frame rate to the display device **150** via the probe transceiver communicating with the display device transceiver over the wireless connection. The acquisition frame rate processor **140** may dynamically update the acquisition frame rate and communicate the updated frame rate to the display device based on changes to the configuration and/or operating condition (e.g., temperature) of the wireless ultrasound probe **104**. In various embodiments, the acquisition frame rate may be transmitted by the acquisition frame rate processor **140** via the probe transceiver over the wireless connection to the display device transceiver of the display device **150** according to the transmission control protocol (TCP) communications protocol.

(23) The display device **150** may be a handheld device (e.g., smart phone, tablet computer, or the like), a laptop computer, a desktop computer, and/or any suitable device having a display system. The display device **150** comprises a user input device **130**, a display signal processor **152**, an image buffer **136**, a display system **134**, an archive **138**, and a display device transceiver. The user input device **130** may be utilized to input patient data, imaging modes, image acquisition and scan parameters, settings, configuration parameters, select protocols and/or templates, select wireless connection state thresholds, and the like. In an exemplary embodiment, the user input device **130** may be operable to configure, manage and/or control operation of one or more components and/or modules in the ultrasound system **100**. In this regard, the user input device **130** may be operable to configure, manage and/or control operation of the transmitter **102**, the ultrasound probe **104**, the transmit beamformer **110**, the receiver **118**, the receive beamformer **120**, the RF processor **124**, the RF/IQ buffer **126**, the user input device **130**, the probe signal processor **132**, the image buffer **136**, the display system **134**, the archive **138**, and/or the probe and display device transceivers. For example, an imaging mode, imaging parameters, imaging presets, settings, and/or the like received via the user input device may be communicated to the wireless ultrasound probe **104** for configuration of the probe **104**. The user input device **130** may include button(s), rotary encoder(s), a touchscreen, motion tracking, voice recognition, a mousing device, keyboard, camera and/or any other device capable of receiving a user directive. In certain embodiments, one or more of the user input devices **130** may be integrated into other components, such as the display system **134**, for example. As an example, the user input device **130** may include a touchscreen display.

(24) The display signal processor **152** may comprise suitable logic, circuitry, interfaces and/or code that may be operable to process the ultrasound image frames received from the wireless ultrasound probe **104** for presentation at the display system **134**. The display signal processor **152** is operable to perform one or more processing operations according to a plurality of selectable ultrasound modalities on the received ultrasound image frames. In an exemplary embodiment, the display signal processor **150** may be operable to perform display processing and/or control processing, among other things. In various embodiments, the processed ultrasound image frames can be

presented at the display system **134** and/or may be stored at the archive **138**. The archive **138** may be a local archive, a Picture Archiving and Communication System (PACS), or any suitable device for storing images and related information.

(25) The display signal processor **152** may be one or more central processing units, graphic processing units, microprocessors, microcontrollers, and/or the like. The display signal processor **152** may be an integrated component, or may be distributed across various locations, for example. In an exemplary embodiment, the display signal processor **152** may comprise an imaging mode processor **160**, a counter processor **170**, and a connection state processor **180** and may be capable of receiving input information from a user input device **130** and/or archive **138**, generating an output displayable by a display system **134**, and manipulating the output in response to input information from a user input device **130**, among other things. The display signal processor **152**, imaging mode processor **160**, counter processor **170**, and connection state processor **180** may be capable of executing any of the method(s) and/or set(s) of instructions discussed herein in accordance with the various embodiments, for example.

(26) The display signal processor **152** may include an imaging mode processor **160** that comprises suitable logic, circuitry, interfaces and/or code that may be operable to receive and/or retrieve an imaging mode, imaging parameters, imaging presets, settings, and/or the like (i.e., wireless ultrasound probe **104** imaging settings) from the user input device **130** and/or the archive **138** for configuration of the wireless ultrasound probe **104**. For example, the imaging mode processor **160** may receive user instructions via the user input device **130** to set or change the wireless ultrasound probe **104** imaging settings. Additionally and/or alternatively, the imaging mode processor **160** may retrieve default and/or stored wireless ultrasound probe **104** imaging settings from the archive **138** and/or any suitable data storage medium. The imaging mode processor **160** may be configured to transmit the wireless ultrasound probe **104** imaging settings via a display device transceiver over a wireless connection to the probe signal processor **132** of the wireless ultrasound probe **104**. The display device transceiver may be configured to transmit the wireless ultrasound probe **104** imaging settings to the wireless ultrasound probe **104** via the wireless connection according to the transmission control protocol (TCP) communications protocol. The wireless ultrasound probe **104** imaging settings may be provided by the probe transceiver to the probe signal processor **132** for controlling configuration and/or operation of the wireless ultrasound probe **104**.

(27) The display signal processor **152** may include a counter processor **170** that comprises suitable logic, circuitry, interfaces and/or code that may be operable to count a number of ultrasound image frames received from the wireless ultrasound probe **104** over an analysis interval length determined by the connection state processor **180** as described below. The counter processor **170** may be configured to provide the number of received ultrasound image frames over each analysis interval length to the connection state processor **180**. The counter processor **170** may be configured to reset the count to zero (0) after each analysis interval length. The counter processor **170** may be configured to reset the count to zero (0) in response to an instruction from the connection state processor **180**. For example, as described below, the connection state processor **180** may instruct the counter processor **170** to reset in response to receiving an updated acquisition frame rate from the wireless ultrasound probe **104**.

(28) The display signal processor **152** may include a connection state processor **180** that comprises suitable logic, circuitry, interfaces and/or code that may be operable to determine a connection state of the wireless connection between the wireless ultrasound probe **104** and the display device **150** and cause the display system **134** to present an indication of the connection state. The connection state processor **180** may be configured to receive the acquisition frame rate from the wireless ultrasound probe **104**. The connection state processor **180** may be configured to determine the at least one wireless connection state threshold. For example, the connection state processor **180** may determine a first threshold for changing from a good connection state to a bad connection state and a second threshold for changing from the bad connection state to the good connection state. In

various embodiments, the first threshold and the second threshold may be different. For example, the first threshold may be a loss of 10% of the ultrasound image frames expected to be received based on the acquisition frame rate to change from a good connection state to a bad connection state and the second threshold may be a loss of 5% of the ultrasound image frames expected to be received based on the acquisition frame rate to change from a bad connection state to a good connection state. The second threshold may be less loss in order to ensure a good connection has been reestablished. The at least one threshold may be a default, selected by a user, and/or determined based on the received acquisition frame rate. As an example, the connection state processor **180** may define the at least one wireless connection state loss threshold to be greater allowed loss for higher frame rates, and less allowed loss for lower frame rates. As another example, the connection state processor **180** may define the at least one wireless connection state loss threshold based on a user input received via the user input device selecting the at least one loss threshold. In various embodiments, default loss thresholds may be pre-defined and stored at archive **138** and/or any suitable data storage medium for retrieval by the connection state processor **180**. The connection state processor **180** may be configured to dynamically determine and/or update the at least one wireless connection state loss threshold in response to receiving the acquisition frame rate and/or an updated acquisition frame rate from the wireless ultrasound probe **104**.

(29) The connection state processor **180** may be configured to determine an analysis interval length based on the received acquisition frame rate and the wireless connection state loss threshold. The interval length is the amount of time the received ultrasound image frames are counted to determine the connection state. In various embodiments, the interval length may be between one (1) second and six (6) seconds but may be lower or higher. The connection state processor **180** may determine longer interval lengths (e.g., 4-6 seconds) for smaller loss thresholds (e.g., 5% loss) and/or smaller acquisition frame rates (e.g., 5-15 frames per second). The connection state processor **180** may determine smaller interval lengths (e.g., 1-3 seconds) for greater loss threshold (e.g., 10% loss) and/or higher acquisition frame rates (e.g., 20-30 frames per second). The connection state processor **180** may determine the interval length in a weighted or non-weighted manner based on the received acquisition frame rate and the wireless connection state loss threshold. The connection state processor **180** may be configured to retrieve the number of received ultrasound image frames counted by the counter processor **170** periodically at the end of the determined interval length. The connection state processor **180** may dynamically determine and/or update the interval length in response to receiving the acquisition frame rate and/or an updated acquisition frame rate from the wireless ultrasound probe **104**. The connection state processor **180** may instruct the counter processor **170** to reset in response to receiving the acquisition frame rate and/or updated acquisition frame rate from the wireless ultrasound probe **104**.

(30) The connection state processor **180** may be configured to determine whether the ultrasound image frames lost exceeds the wireless connection state loss threshold based on the acquisition frame rate and the interval length. For example, if the current wireless connection state is a good state, the acquisition frame rate is 25 frames per second, the analysis interval length is 1.0 seconds, the loss threshold is 10% of the ultrasound image frames expected to be received, and the number of counted received ultrasound image frames over the analysis interval length is 20 frames, the number of ultrasound image frames lost (i.e., 25 expected-20 received=5 frames) would exceed the loss threshold of 2.5 frames (i.e., 10% of 25 frames/second*1.0 seconds). Accordingly, the connection state processor **180** would change the connection state from a good state to a bad state.

(31) The connection state processor **180** may be configured to change or maintain the connection state based on the current connection state and the determination of whether the ultrasound image frames lost exceed the wireless connection state loss threshold. For example, the connection state processor **180** may update the connection state from a good state to a bad state when the ultrasound image frames lost exceed the wireless connection state loss threshold. The connection state processor **180** may update the connection state from a bad state to a good state when the ultrasound

image frames lost do not exceed the wireless connection state loss threshold. The connection state processor **180** may maintain the connection state in a good state when the ultrasound image frames lost do not exceed the wireless connection state loss threshold. The connection state processor may maintain the connection state in a bad state when the ultrasound image frames lost exceeds the wireless connection state loss threshold. The connection state processor **180** may be configured to cause the display system **134** of the display device **150** to present the connection state. For example, the connection state may be represented by an icon, color-coding, text, and/or the like. The connection state may be continuously presented at the display system **134** and/or alerts regarding a change in the connection state may be presented at the display system **134**.

(32) FIG. 2 illustrates a screenshot of an exemplary ultrasound image display **200** having an indication **202** of a good connection status, in accordance with various embodiments. Referring to FIG. 2, the ultrasound image display **200** includes an exemplary wireless icon **202** that may be color-coded green to represent a good connection status. The indication **202** of the good connection status is shown presented at a top of the ultrasound image display **200** but could be presented at any suitable position on the ultrasound image display **200**. The indication **202** of the good connection status may be continuously presented and/or dynamically updated to provide a user feedback regarding the connection status, such that the user is aware whether the acquired ultrasound image frames are substantially present or if an unacceptable number of ultrasound image frames may be missing.

(33) FIG. 3 illustrates a screenshot of an exemplary ultrasound image display **200** having an indication **204** of a bad connection status, in accordance with various embodiments. Referring to FIG. 3, the ultrasound image display **200** includes an exemplary wireless icon **204**, which may be color-coded grey, with an exclamation point, which may be color-coded yellow or red, to represent a bad connection status. The indication **204** of the bad connection status is shown presented at a top of the ultrasound image display **200** but could be presented at any suitable position on the ultrasound image display **200**. The indication **204** of the bad connection status may be continuously presented and/or dynamically updated to provide a user feedback regarding the connection status, such that the user is aware whether the acquired ultrasound image frames are substantially present or if an unacceptable number of ultrasound image frames may be missing.

(34) Referring again to FIG. 1, the image buffer **136** of the display device **150** is included for storing processed frames of received ultrasound image frames that are not scheduled to be displayed immediately. Preferably, the image buffer **136** is of sufficient capacity to store at least several minutes' worth of ultrasound image frames. The ultrasound image frames are stored in a manner to facilitate retrieval thereof according to its order or time of acquisition. The image buffer **136** may be embodied as any known data storage medium.

(35) The display system **134** may be any device capable of communicating visual information to a user. For example, a display system **134** may include a liquid crystal display, a light emitting diode display, and/or any suitable display or displays. The display system **134** can be operable to present ultrasound image frames, a connection status **202**, **204**, and/or any suitable information.

(36) The archive **138** may be one or more computer-readable memories integrated with the display device **150** and/or communicatively coupled (e.g., over a network) to the display device **150**, such as a Picture Archiving and Communication System (PACS), a server, a hard disk, floppy disk, CD, CD-ROM, DVD, compact storage, flash memory, random access memory, read-only memory, electrically erasable and programmable read-only memory and/or any suitable memory. The archive **138** may include databases, libraries, sets of information, or other storage accessed by and/or incorporated with the display device signal processor **152**, for example. The archive **138** may be able to store data temporarily or permanently, for example. The archive **138** may be capable of storing medical image data, data generated by the display device signal processor **152**, the probe signal processor **132**, and/or instructions readable by the signal processor(s) **132**, **152**, among other things. In various embodiments, the archive **138** stores ultrasound images, wireless ultrasound

probe settings, received acquisition frame rates, wireless connection state loss thresholds, analysis interval lengths, instructions for determining loss thresholds, instructions for determining analysis interval lengths, instructions for determining whether lost frames exceed the loss threshold, and/or instructions for presenting a connection status, for example.

(37) Components of the ultrasound system **100** may be implemented in software, hardware, firmware, and/or the like. The various components of the ultrasound system **100** may be communicatively linked. Components of the ultrasound system **100** may be implemented separately and/or integrated in various forms. For example, the display system **134** and the user input device **130** may be integrated as a touchscreen display.

(38) FIG. **4** is a flow chart **300** illustrating exemplary steps **302-330** that may be utilized for providing feedback **202**, **204** to a user of a wireless ultrasound probe **104** regarding a wireless link quality between the ultrasound probe **104** and a display device **150**, in accordance with various embodiments. Referring to FIG. **4**, there is shown a flow chart **300** comprising exemplary steps **302** through **330**. Certain embodiments may omit one or more of the steps, and/or perform the steps in a different order than the order listed, and/or combine certain of the steps discussed below. For example, some steps may not be performed in certain embodiments. As a further example, certain steps may be performed in a different temporal order, including simultaneously, than listed below.

(39) At step **302**, a display device **150** transmits wireless ultrasound probe imaging settings to a wireless ultrasound probe **104**. For example, an imaging mode processor **160** of the display device signal processor **152** of the display device may be operable to receive and/or retrieve an imaging mode, imaging parameters, imaging presets, settings, and/or the like (i.e., wireless ultrasound probe imaging settings) from a user input device **130** and/or the archive **138** for configuration of the wireless ultrasound probe **104**. The wireless ultrasound probe imaging settings may be user selected settings and/or default settings. The imaging mode processor **160** may be configured to transmit the wireless ultrasound probe imaging settings via a display device transceiver over a wireless connection to the probe signal processor **132** of the wireless ultrasound probe **104** according to the transmission control protocol (TCP) communications protocol. The wireless ultrasound probe imaging settings may be provided by the probe transceiver to the probe signal processor **132** for controlling configuration and/or operation of the wireless ultrasound probe **104**.

(40) At step **304**, the wireless ultrasound probe **104** determines an acquisition frame rate based on the wireless ultrasound probe imaging settings received from the display device **150**. For example, an acquisition frame rate processor **140** of a probe signal processor **132** of the wireless ultrasound probe **104** may be operable to determine an acquisition frame rate of the wireless ultrasound probe **104**. The probe signal processor **132** may receive an imaging mode, imaging parameters, imaging presets, settings, and/or the like from the display device **150** via the wireless connection (e.g., using TCP) to configure the wireless ultrasound probe **104** for ultrasound image acquisition at step **302**. The acquisition frame rate processor **140** may be configured to determine the acquisition frame rate based on the configuration and operating condition of the wireless ultrasound probe **104**. The acquisition frame rate processor **140** may dynamically update the acquisition frame rate based on changes to the configuration and/or operating condition (e.g., temperature) of the wireless ultrasound probe **104**.

(41) At step **306**, the wireless ultrasound probe **104** transmits the determined acquisition frame rate to the display device **150**. For example, the acquisition frame rate processor **140** may be configured to transmit the determined acquisition frame rate to the display device **150** via the probe transceiver communicating with the display device transceiver over the wireless connection according to the transmission control protocol (TCP) communications protocol.

(42) At step **308**, the display device **150** determines a wireless connection state loss threshold. For example, a connection state processor **180** of the display device signal processor **152** of the display device **150** may be configured to determine the at least one wireless connection state loss threshold. The connection state processor **180** may determine a first threshold for changing from a good

connection state to a bad connection state and a second threshold for changing from the bad connection state to the good connection state. In various embodiments, the first threshold and the second threshold may be different. The at least one wireless connection state loss threshold may be a default, selected by a user, and/or determined based on the received acquisition frame rate (e.g., greater loss threshold for higher acquisition frame rates). The connection state processor **180** may be configured to dynamically determine and/or update the at least one wireless connection state loss threshold in response to receiving the acquisition frame rate and/or an updated acquisition frame rate from the wireless ultrasound probe **104**.

(43) At step **310**, the display device **150** determines an analysis interval length based on the acquisition frame rate and the wireless connection state loss threshold. For example, the analysis interval length is the amount of time the received ultrasound image frames are counted to determine the connection state. In various embodiments, the interval length may be between one (1) second and six (6) seconds but may be lower or higher. The connection state processor **180** may determine longer interval lengths (e.g., 4-6 seconds) for smaller loss thresholds (e.g., 5% loss) and/or smaller acquisition frame rates (e.g., 5-15 frames per second). The connection state processor **180** may determine smaller interval lengths (e.g., 1-3 seconds) for greater loss threshold (e.g., 10% loss) and/or higher acquisition frame rates (e.g., 20-30 frames per second). The connection state processor **180** may determine the interval length in a weighted or non-weighted manner based on the received acquisition frame rate and the wireless connection state loss threshold. The connection state processor **180** may dynamically determine and/or update the interval length in response to receiving the acquisition frame rate and/or an updated acquisition frame rate from the wireless ultrasound probe **104**.

(44) At step **312**, the wireless ultrasound probe **104** acquires ultrasound image data and communicates ultrasound image frames to the display device **150**. For example, a wireless ultrasound probe **104** in the ultrasound system **100** may be operable to continuously acquire ultrasound scan data at an acquisition frame rate that is suitable for the imaging situation in question. Typical acquisition frame rates range from 5-30 frames per second but may be lower or higher. The acquired ultrasound scan data processed as ultrasound image frames by the probe signal processor **132** may be wirelessly transmitted to the display device **150** for further display processing and display on the display system **134** at a display-rate that can be the same as the frame rate, or slower or faster. The wireless connection between the wireless probe **104** and the display device **150** may be Wi-Fi or any suitable wireless connection. In various embodiments, a probe transceiver may be configured to transmit ultrasound image frames from the probe signal processor **132** to a display device transceiver of the display device **150** via the wireless connection according to the user datagram protocol (UDP) communications protocol.

(45) At step **314**, the display device **150** counts the received ultrasound image frames. For example, a counter processor **170** of the display device signal processor **152** may be operable to count a number of ultrasound image frames received from the wireless ultrasound probe **104** over an analysis interval length determined by the connection state processor **180** at step **310**. The counter processor **170** may be configured to provide the number of received ultrasound image frames over each analysis interval length to the connection state processor **180**. The counter processor **170** may be configured to reset the count to zero (0) after each analysis interval length. The counter processor **170** may be configured to reset the count to zero (0) in response to an instruction from the connection state processor **180**, such as in response to receiving an updated acquisition frame rate from the wireless ultrasound probe **104**.

(46) At step **316**, if the connection state between the wireless ultrasound probe **104** and the display device is a good connection state, the method **300** proceeds to step **318**. If the connection state between the wireless ultrasound probe **104** and the display device is a bad connection state, the method **300** proceeds to step **324**. Initially, the connection state may begin in either of the good or bad connection states by default, user selection, or any suitable connection state analysis process.

(47) At step **318**, the display device **150** determines whether the lost ultrasound image frames exceed a first wireless connection state loss threshold. For example, the connection state processor **180** may determine a number of frames lost by subtracting the counted received ultrasound image frames as determined by the counter processor **170** from the expected number of received ultrasound image frames determined based on the acquisition frame rate and the analysis interval length (e.g., 25 frames/second over a 2.0 second analysis interval length would equal 50 expected ultrasound image frames). The connection state processor **180** compares the number of frames lost to a threshold loss number determined by multiplying the determined loss threshold to the acquisition rate, and multiplying the result by the analysis interval length (e.g., 10% loss threshold*25 frames/second*2.0 seconds=5 frames). If the number of lost frames exceeds the threshold loss number, the connection state processor **180** determines that the lost ultrasound image frames exceed a first wireless connection state loss threshold. If the number of lost frames is less than the threshold loss number, the connection state processor **180** determines that the lost ultrasound image frames do not exceed a first wireless connection state loss threshold. In various embodiments, the first wireless connection state loss threshold may be a threshold specific to the good connection state and may be different from a second wireless connection state loss threshold specific to the bad connection state.

(48) At step **320**, if the number of lost frames exceeds the first wireless connection state loss threshold, the method proceeds to step **322**. If the number of lost frames does not exceed the first wireless connection state loss threshold, the method proceeds to step **330**.

(49) At step **322**, the display device **150** changes the connection state from a good connection state to a bad connection state and updates the displayed state. For example, the connection state processor **180** may update the connection state from a good state to a bad state when the ultrasound image frames lost exceeds the wireless connection state loss threshold as determined at step **318**. The connection state may be represented by an icon, color-coding, text, and/or the like. The connection state may change from the good connection state to the bad connection state and be continuously presented at the display system **134** such that the user is aware that an unacceptable number of ultrasound image frames may be missing.

(50) At step **324**, the display device **150** determines whether the lost ultrasound image frames exceeds a second wireless connection state loss threshold. For example, the connection state processor **180** may determine a number of frames lost by subtracting the counted received ultrasound image frames as determined by the counter processor **170** from the expected number of received ultrasound image frames determined based on the acquisition frame rate and the analysis interval length. The connection state processor **180** compares the number of frames lost to a threshold loss number determined by multiplying the determined loss threshold to the acquisition rate, and multiplying the result by the analysis interval length. If the number of lost frames exceeds the threshold loss number, the connection state processor **180** determines that the lost ultrasound image frames exceed a second wireless connection state loss threshold. If the number of lost frames is less than the threshold loss number, the connection state processor **180** determines that the lost ultrasound image frames do not exceed a second wireless connection state loss threshold. In various embodiments, the second wireless connection state loss threshold may be a threshold specific to the bad connection state and may be different from a first wireless connection state loss threshold specific to the good connection state used in step **318**. For example, the first threshold may be a loss of 10% (or any suitable percentage) of the ultrasound image frames expected to be received based on the acquisition frame rate to change from a good connection state to a bad connection state and the second threshold may be a loss of 5% (or any suitable percentage) of the ultrasound image frames expected to be received based on the acquisition frame rate to change from a bad connection state to a good connection state. The second threshold may be a lower loss threshold in order to ensure a good connection has been reestablished.

(51) At step **326**, if the number of lost frames exceeds the second wireless connection state loss

threshold, the method proceeds to step **330**. If the number of lost frames does not exceed the first wireless connection state loss threshold, the method proceeds to step **328**.

(52) At step **328**, the display device **150** changes the connection state from a bad connection state to a good connection state and updates the displayed state. For example, the connection state processor **180** may update the connection state from a bad state to a good state when the ultrasound image frames lost do not exceed the wireless connection state loss threshold as determined at step **324**. The connection state may be represented by an icon, color-coding, text, and/or the like. The connection state may change from the bad connection state to the good connection state and be continuously presented at the display system **134** such that the user is aware that the acquired ultrasound image frames are substantially present.

(53) At step **330**, the display device **150** maintains and continues to present the connection state. For example, the connection state processor **180** may maintain the connection state in a good state when the ultrasound image frames lost do not exceed the wireless connection state loss threshold. The connection state processor may maintain the connection state in a bad state when the ultrasound image frames lost exceed the wireless connection state loss threshold. The connection state processor **180** may be configured to cause the display system **134** of the display device **150** to present the connection state. For example, the connection state may be represented by an icon, color-coding, text, and/or the like. The connection state may be continuously presented at the display system **134** such that the user is aware whether the acquired ultrasound image frames are substantially present or if an unacceptable number of ultrasound image frames may be missing.

(54) Steps **316** through **330** may be repeated periodically. For example, the connection state processor **180** may be configured to retrieve the number of received ultrasound image frames counted by the counter processor **170** at step **314** periodically at the end of the interval length determined at step **310**.

(55) Aspects of the present disclosure provide a method **300** and system **100** for providing feedback **202**, **204** to a user of a wireless ultrasound probe **104** regarding a wireless link quality between the ultrasound probe **104** and a display device **150**. The method **300** may comprise receiving **306**, by at least one display processor **152**, **160** of a display device **150**, an acquisition frame rate from a wireless ultrasound probe **104**. The method **300** may comprise determining **310**, by the at least one display processor **152**, **180**, a first analysis interval length based on the acquisition frame rate and a first wireless connection state loss threshold. The method **300** may comprise receiving **312** and counting **314**, by the at least one display processor **152**, **170**, ultrasound image frames wirelessly transmitted to the display device **150** from the wireless ultrasound probe **104** during the first analysis interval length. The method **300** may comprise determining **318**, by the at least one display processor **152**, **180**, that a first number of lost ultrasound frames during the first analysis interval length exceeds the first wireless connection state loss threshold. The method **300** may comprise causing **322**, by the at least one display processor **152**, **180**, a display system **134** of the display device **150** to present a bad connection state identifier **204**.

(56) In a representative embodiment, the method **300** may comprise determining **310**, by the at least one display processor **152**, **180**, a second analysis interval length based on the acquisition frame rate and a second wireless connection state loss threshold. The method **300** may comprise receiving **312** and counting **314**, by the at least one display processor **152**, **170**, the ultrasound image frames wirelessly transmitted to the display device **150** from the wireless ultrasound probe **104** during the second analysis interval length. The method **300** may comprise determining **324**, by the at least one display processor **152**, **180**, that a second number of lost ultrasound frames during the second analysis interval length exceeds the second wireless connection state loss threshold. The method **300** may comprise causing **330**, by the at least one display processor **152**, **180**, the display system **134** to continue presenting the bad connection state identifier **204**. In an exemplary embodiment, the method **300** may comprise determining **310**, by the at least one display processor **152**, **180**, a second analysis interval length based on the acquisition frame rate and a second

wireless connection state loss threshold. The method **300** may comprise receiving **312** and counting **314**, by the at least one display processor **152**, **170**, the ultrasound image frames wirelessly transmitted to the display device **150** from the wireless ultrasound probe **104** during the second analysis interval length. The method **300** may comprise determining **324**, by the at least one display processor **152**, **180**, that a second number of lost ultrasound frames during the second analysis interval length does not exceed the second wireless connection state loss threshold. The method **300** may comprise causing **328**, by the at least one display processor **152**, **180**, the display system **134** to change from presenting the bad connection state identifier **204** to presenting a good connection state identifier **202**. In various embodiments, the first wireless connection state loss threshold corresponds with a good connection state and the second wireless connection state loss threshold corresponds with a bad connection state. In certain embodiments, the first wireless connection state loss threshold is larger than the second wireless connection state loss threshold. In a representative embodiment, the first wireless connection state loss threshold is based on the acquisition frame rate. In an exemplary embodiment, the method **300** may comprise communicating **302**, by the display device **150**, wireless ultrasound probe imaging settings to the wireless ultrasound probe **104**. The acquisition frame rate may be determined by the wireless ultrasound probe **104** based on the wireless ultrasound probe imaging settings.

(57) Various embodiments provide an ultrasound system **100** for providing feedback **202**, **204** to a user of a wireless ultrasound probe **104** regarding a wireless link quality between the ultrasound probe **104** and a display device **150**. The ultrasound system **100** may comprise a wireless ultrasound probe **104** and a display device **150**, and at least one processor **132**, **140**, **152**, **160**, **170**, **180**. The wireless ultrasound probe **104** may be operable to wirelessly transmit an acquisition frame rate to a display device **150**. The wireless ultrasound probe **104** may be operable to wirelessly transmit ultrasound image frames to the display device **150**. The display device **150** may comprise at least one display processor **152**, **160**, **170**, **180** and a display system **134**. The at least one display processor **152**, **180** may be configured to determine an analysis interval length based on the acquisition frame rate and a wireless connection state loss threshold. The at least one display processor **152**, **170** may be configured to count the ultrasound image frames received from the wireless ultrasound probe **104** during the analysis interval length. The at least one display processor **150**, **180** may be configured to determine whether a number of lost ultrasound frames during the analysis interval length exceeds the wireless connection state loss threshold. The display system **134** may be configured to present a bad connection state identifier when the number of lost ultrasound frames during the analysis interval length exceeds the wireless connection state loss threshold. The display system **134** may be configured to present a good connection state identifier when the number of lost ultrasound frames during the analysis interval length does not exceed the wireless connection state loss threshold.

(58) In an exemplary embodiment, the ultrasound image frames are wirelessly transmitted to the display device **150** according to a user datagram protocol (UDP) communications protocol. In various embodiments, the display device **150** is configured to wirelessly transmit wireless ultrasound probe imaging settings to the wireless ultrasound probe **104**. The wireless ultrasound probe **104** may comprise at least one probe processor **132**, **140** configured to determine the acquisition frame rate based on the wireless ultrasound probe imaging settings. In certain embodiments, the wireless ultrasound probe imaging settings are wirelessly transmitted from the display device **150** to the wireless ultrasound probe **104** according to a transmission control protocol (TCP) communications protocol. In a representative embodiment, the acquisition frame rate is wirelessly transmitted from the wireless ultrasound probe **104** to the display device **150** according to a transmission control protocol (TCP) communications protocol. In an exemplary embodiment, the wireless connection state loss threshold is based on a current state of a wireless connection between the wireless ultrasound probe **104** and the display device **150**. In various embodiments, the wireless connection state loss threshold is based on the acquisition frame rate. In

certain embodiments, a wireless connection between the wireless ultrasound probe **104** and the display device **150** is a Wi-Fi connection.

(59) Certain embodiments provide a non-transitory computer readable medium having stored thereon, a computer program having at least one code section. The at least one code section is executable by a machine for causing the machine to perform steps **300**. The steps **300** may comprise receiving **306** an acquisition frame rate from a wireless ultrasound probe **104** at a display device **150**. The steps **300** may comprise determining **310** a first analysis interval length based on the acquisition frame rate and a first wireless connection state loss threshold. The steps **300** may comprise receiving **312** and counting **314** ultrasound image frames wirelessly transmitted to the display device **150** from the wireless ultrasound probe **104** during the first analysis interval length. The steps **300** may comprise determining **318** that a first number of lost ultrasound frames during the first analysis interval length exceeds the first wireless connection state loss threshold. The steps **300** may comprise causing **322** a display system **134** of the display device **150** to present a bad connection state identifier **204**.

(60) In various embodiments, the steps **300** may comprise determining **310** a second analysis interval length based on the acquisition frame rate and a second wireless connection state loss threshold. The steps **300** may comprise receiving **312** and counting **314** the ultrasound image frames wirelessly transmitted to the display device **150** from the wireless ultrasound probe **104** during the second analysis interval length. The steps **300** may comprise determining **324** that a second number of lost ultrasound frames during the second analysis interval length exceeds the second wireless connection state loss threshold. The steps **300** may comprise causing **330** the display system **134** to continue presenting the bad connection state identifier **204**. In certain embodiments, the steps **300** may comprise determining **310** a second analysis interval length based on the acquisition frame rate and a second wireless connection state loss threshold. The steps **300** may comprise receiving **312** and counting **314** the ultrasound image frames wirelessly transmitted to the display device **150** from the wireless ultrasound probe **104** during the second analysis interval length. The steps **300** may comprise determining **324** that a second number of lost ultrasound frames during the second analysis interval length does not exceed the second wireless connection state loss threshold. The steps **300** may comprise causing **328** the display system **134** to change from presenting the bad connection state identifier **204** to presenting a good connection state identifier **202**. In a representative embodiment, the first wireless connection state loss threshold corresponds with a good connection state and the second wireless connection state loss threshold corresponds with a bad connection state. The first wireless connection state loss threshold may be larger than the second wireless connection state loss threshold. In an exemplary embodiment, the first wireless connection state loss threshold is based on the acquisition frame rate.

(61) As utilized herein the term “circuitry” refers to physical electronic components (i.e. hardware) and any software and/or firmware (“code”) which may configure the hardware, be executed by the hardware, and or otherwise be associated with the hardware. As used herein, for example, a particular processor and memory may comprise a first “circuit” when executing a first one or more lines of code and may comprise a second “circuit” when executing a second one or more lines of code. As utilized herein, “and/or” means any one or more of the items in the list joined by “and/or”. As an example, “x and/or y” means any element of the three-element set {(x), (y), (x, y)}. As another example, “x, y, and/or z” means any element of the seven-element set {(x), (y), (z), (x, y), (x, z), (y, z), (x, y, z)}. As utilized herein, the term “exemplary” means serving as a non-limiting example, instance, or illustration. As utilized herein, the terms “e.g.,” and “for example” set off lists of one or more non-limiting examples, instances, or illustrations. As utilized herein, circuitry is “operable” and/or “configured” to perform a function whenever the circuitry comprises the necessary hardware and code (if any is necessary) to perform the function, regardless of whether performance of the function is disabled, or not enabled, by some user-configurable setting.

(62) Other embodiments may provide a computer readable device and/or a non-transitory computer readable medium, and/or a machine readable device and/or a non-transitory machine readable medium, having stored thereon, a machine code and/or a computer program having at least one code section executable by a machine and/or a computer, thereby causing the machine and/or computer to perform the steps as described herein for providing feedback to a user of a wireless ultrasound probe regarding a wireless link quality between the ultrasound probe and a display device.

(63) Accordingly, the present disclosure may be realized in hardware, software, or a combination of hardware and software. The present disclosure may be realized in a centralized fashion in at least one computer system, or in a distributed fashion where different elements are spread across several interconnected computer systems. Any kind of computer system or other apparatus adapted for carrying out the methods described herein is suited.

(64) Various embodiments may also be embedded in a computer program product, which comprises all the features enabling the implementation of the methods described herein, and which when loaded in a computer system is able to carry out these methods. Computer program in the present context means any expression, in any language, code or notation, of a set of instructions intended to cause a system having an information processing capability to perform a particular function either directly or after either or both of the following: a) conversion to another language, code or notation; b) reproduction in a different material form.

(65) While the present disclosure has been described with reference to certain embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted without departing from the scope of the present disclosure. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from its scope. Therefore, it is intended that the present disclosure not be limited to the particular embodiment disclosed, but that the present disclosure will include all embodiments falling within the scope of the appended claims.

Claims

1. A method, comprising: receiving, by at least one display processor of a display device, an acquisition frame rate from a wireless ultrasound probe; determining, by the at least one display processor, a first analysis interval length based on the acquisition frame rate and a first wireless connection state loss threshold; receiving and counting, by the at least one display processor, ultrasound image frames wirelessly transmitted to the display device from the wireless ultrasound probe during the first analysis interval length; determining, by the at least one display processor, that a first number of lost ultrasound frames during the first analysis interval length exceeds the first wireless connection state loss threshold; and causing, by the at least one display processor, a display system of the display device to present a bad connection state identifier.
2. The method of claim 1, further comprising: determining, by the at least one display processor, a second analysis interval length based on the acquisition frame rate and a second wireless connection state loss threshold; receiving and counting, by the at least one display processor, the ultrasound image frames wirelessly transmitted to the display device from the wireless ultrasound probe during the second analysis interval length; determining, by the at least one display processor, that a second number of lost ultrasound frames during the second analysis interval length exceeds the second wireless connection state loss threshold; and causing, by the at least one display processor, the display system to continue presenting the bad connection state identifier.
3. The method of claim 1, further comprising: determining, by the at least one display processor, a second analysis interval length based on the acquisition frame rate and a second wireless connection state loss threshold; receiving and counting, by the at least one display processor, the ultrasound image frames wirelessly transmitted to the display device from the wireless ultrasound

probe during the second analysis interval length; determining, by the at least one display processor, that a second number of lost ultrasound frames during the second analysis interval length does not exceed the second wireless connection state loss threshold; and causing, by the at least one display processor, the display system to change from presenting the bad connection state identifier to presenting a good connection state identifier.

4. The method of claim 3, wherein the first wireless connection state loss threshold corresponds with a good connection state and the second wireless connection state loss threshold corresponds with a bad connection state.

5. The method of claim 4, wherein the first wireless connection state loss threshold is larger than the second wireless connection state loss threshold.

6. The method of claim 1, wherein the first wireless connection state loss threshold is based on the acquisition frame rate.

7. The method of claim 1, comprising: communicating, by the display device, wireless ultrasound probe imaging settings to the wireless ultrasound probe, wherein the acquisition frame rate is determined by the wireless ultrasound probe based on the wireless ultrasound probe imaging settings.

8. An ultrasound system, comprising: a wireless ultrasound probe operable to: wirelessly transmit an acquisition frame rate to a display device; and wirelessly transmit ultrasound image frames to the display device; and the display device comprising: at least one display processor configured to: determine an analysis interval length based on the acquisition frame rate and a wireless connection state loss threshold; count the ultrasound image frames received from the wireless ultrasound probe during the analysis interval length; and determine whether a number of lost ultrasound frames during the analysis interval length exceeds the wireless connection state loss threshold; and a display system configured to: present a bad connection state identifier when the number of lost ultrasound frames during the analysis interval length exceeds the wireless connection state loss threshold; and present a good connection state identifier when the number of lost ultrasound frames during the analysis interval length does not exceed the wireless connection state loss threshold.

9. The system of claim 8, wherein the ultrasound image frames are wirelessly transmitted to the display device according to a user datagram protocol (UDP) communications protocol.

10. The system of claim 8, wherein: the display device is configured to wirelessly transmit wireless ultrasound probe imaging settings to the wireless ultrasound probe, and the wireless ultrasound probe comprises at least one probe processor configured to determine the acquisition frame rate based on the wireless ultrasound probe imaging settings.

11. The system of claim 10, wherein the wireless ultrasound probe imaging settings are wirelessly transmitted from the display device to the wireless ultrasound probe according to a transmission control protocol (TCP) communications protocol.

12. The system of claim 8, wherein the acquisition frame rate is wirelessly transmitted from the wireless ultrasound probe to the display device according to a transmission control protocol (TCP) communications protocol.

13. The system of claim 8, wherein the wireless connection state loss threshold is based on a current state of a wireless connection between the wireless ultrasound probe and the display device.

14. The system of claim 8, wherein the wireless connection state loss threshold is based on the acquisition frame rate.

15. The system of claim 8, wherein a wireless connection between the wireless ultrasound probe and the display device is a Wi-Fi connection.

16. A non-transitory computer readable medium having stored thereon, a computer program having at least one code section, the at least one code section being executable by a machine for causing the machine to perform steps comprising: receiving an acquisition frame rate from a wireless ultrasound probe at a display device; determining a first analysis interval length based on the acquisition frame rate and a first wireless connection state loss threshold; receiving and counting

ultrasound image frames wirelessly transmitted to the display device from the wireless ultrasound probe during the first analysis interval length; determining that a first number of lost ultrasound frames during the first analysis interval length exceeds the first wireless connection state loss threshold; and causing a display system of the display device to present a bad connection state identifier.

17. The non-transitory computer readable medium of claim 16, further comprising: determining a second analysis interval length based on the acquisition frame rate and a second wireless connection state loss threshold; receiving and counting the ultrasound image frames wirelessly transmitted to the display device from the wireless ultrasound probe during the second analysis interval length; determining that a second number of lost ultrasound frames during the second analysis interval length exceeds the second wireless connection state loss threshold; and causing the display system to continue presenting the bad connection state identifier.

18. The non-transitory computer readable medium of claim 16, further comprising: determining a second analysis interval length based on the acquisition frame rate and a second wireless connection state loss threshold; receiving and counting the ultrasound image frames wirelessly transmitted to the display device from the wireless ultrasound probe during the second analysis interval length; determining that a second number of lost ultrasound frames during the second analysis interval length does not exceed the second wireless connection state loss threshold; and causing the display system to change from presenting the bad connection state identifier to presenting a good connection state identifier.

19. The non-transitory computer readable medium of claim 18, wherein: the first wireless connection state loss threshold corresponds with a good connection state, the second wireless connection state loss threshold corresponds with a bad connection state, and the first wireless connection state loss threshold is larger than the second wireless connection state loss threshold.

20. The non-transitory computer readable medium of claim 16, wherein the first wireless connection state loss threshold is based on the acquisition frame rate.
